

Multiple Choice (10 marks)

1. Failed sessions allow brute-force attacks on access credentials. This type of attacks are done in which layer of the OSI model?
 - (a) Physical layer
 - (b) Data-link Layer
 - (c) Session layer
 - (d) Presentation layer
2. SHA-1 has a message digest of
 - (a) 160 bits
 - (b) 512 bits
 - (c) 628 bits
 - (d) 820 bits
3. Exploitation of the Heartbleed bug permits
 - (a) overwriting cryptographic keys in memory
 - (b) a kind of code injection
 - (c) a read outside bounds of a buffer
 - (d) a format string attack
4. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
 - (a) Joined
 - (b) Authenticated
 - (c) Submitted
 - (d) Shared
5. How is IP address spoofing detected?
 - (a) Installing and configuring a IDS that can read the IP header
 - (b) Comparing the TTL values of the actual and spoofed addresses
 - (c) Implementing a firewall to the network
 - (d) Identify all TCP sessions that are initiated but does not complete successfully

True / False (10 marks)

Answer True or False

6. True or False: DaCryptic is a widely recognized remote Trojan that primarily targets financial institutions.
7. True or False: The stack is used for storing dynamically linked libraries required by a program at runtime.
8. True or False: Applications developed using C and C++ are commonly susceptible to buffer-overflow vulnerabilities due to their memory management practices.
9. True or False: All stream ciphers are capable of achieving perfect secrecy under certain conditions.
10. True or False: Wireless access refers to the methods used to connect devices to a network, rather than the security of that network.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the sum of first even and odd number of a given list.
12. Write a python function to find the minimum number of subsets with distinct elements.

End of Paper